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Supplemental Material

Prenatal Exposure to PM_{2.5} and Cardiac Vagal Tone during Infancy: Findings from a Multiethnic Birth Cohort

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Table S1. Mean±SD infant activity levels at each SFP-R episode stratified by infant sex and number of sequences completed.

		Infant sex		SFP-R sequences completed	
SFP-R episode	All children (n=237)	Girls (n=108)	Boys (n=129)	Both (n=185)	First only (n=52)
Play	80.6±40.1	79.7±40.1	81.3±40.2	79.4±38.4	84.8±45.7
Still-Face 1	158.6±34.7	154.6±30.9	161.9±37.3	157.5±33.5	162.3±38.6
Reunion 1	98.5±43.8	95.5±40.3	101.1±56.6	89.9±39.3	129.2±45.7
Still-Face 2	159.1±34.7	157.8±37.3	160.3±38.0	159.1±37.6	
Reunion 2	96.5±45.0	95.8±44.9	97.2±45.4	96.5±45.0	
Abbreviations: SFP-R: Repeated Still-Face Paradigm					

Table S2. Comparison of PRISM study participant characteristics (continuous) stratified by inclusion vs. exclusion from the current analysis and by whether the infant completed only the first vs. both SFP-R sequences.

	Included vs. excluded maternal-child pairs					SFP-R sequences ^b completed				
	Included		Excluded			Both		First only		
	Mean±SD	N	Mean±SD	N	P-value ^c	Mean±SD	N	Mean±SD	N	P-value ^c
Prenatal PM _{2.5} (µg/m ³)	8.3±0.9	237	8.3±0.8	153	0.32	8.3±0.9	185	8.4±0.9	52	0.63
Maternal age at enrollment (years)	30.8±5.5	237	31.5±5.4	179	0.22	31.1±5.5	185	30.0±5.3	52	0.22
Maternal stress ^a	9.6±10.8	237	9.0±8.6	164	0.58	9.4±11.3	185	10.0±9.3	52	0.73
Gestational age (weeks)	39.0±1.8	237	38.9±1.6	129	0.85	39.0±1.7	185	38.9±2.1	52	0.86
Birthweight (kg)	3.2±0.6	237	3.2±0.5	126	0.88	3.2±0.5	185	3.2±0.7	52	0.84

Abbreviations: PM_{2.5}: fine particulate matter; PRISM: PRogramming of Intergenerational Stress Mechanisms; SFP-R: Repeated Still-Face Paradigm

^aAssessed using the Life Stressor Checklist- Revised

^bSequence 1 includes the Play, Still-Face 1 and Reunion 1 episodes; Sequence 2 includes the Still-Face 2 and Reunion 2 episodes.

^cP-values are from t-tests comparing group means.

Table S3. Comparison of PRISM study participant characteristics (categorical) stratified by inclusion vs. exclusion from the current analysis and by whether the infant completed only the first vs. both SFP-R sequences. P-values are from chi-square tests of independence.

	Included vs. excluded maternal-child pairs					SFP-R sequences ^b completed				
	Included		Excluded			Both		First only		
	N (%)	N	N (%)	N	P-value	N (%)	N	N (%)	N	P-value
Maternal race/ethnicity		237		176	0.60		185		52	0.95
White	80 (34)		68 (39)			63 (34)		17 (33)		
Black	61 (26)		46 (26)			46 (25)		15 (29)		
Hispanic	77 (32)		47 (27)			61 (33)		16 (31)		
Other/mixed	19 (8)		15 (9)			15 (8)		4 (8)		
Maternal high school degree		234		174	0.02		184		50	0.56
No	54 (23)		24 (14)			44 (24)		10 (20)		
Yes	180 (77)		150 (86)			140 (76)		40 (80)		
Material hardship		237		176	0.40		185		52	0.47
Not likely	163 (69)		126 (72)			128 (69)		35 (67)		
Somewhat likely	48 (20)		27 (15)			39 (21)		9 (17)		
Likely, very likely or extremely likely	26 (11)		23 (13)			18 (10)		8 (15)		
Tobacco smoke exposure ^a		237		176	0.69		185		52	0.03
No	200 (84)		151 (86)			161 (87)		39 (75)		
Yes	37 (16)		25 (14)			24 (13)		13 (25)		
Infant sex		237		128	0.17		185		52	0.59
Male	129 (54)		60 (47)			99 (54)		30 (58)		
Female	108 (46)		68 (53)			86 (46)		22 (46)		

Abbreviations: PRISM: PROgramming of Intergenerational Stress Mechanisms; SFP-R: Repeated Still-Face Paradigm

^aMaternal self-report of smoking or exposure to environmental tobacco smoke for ≥1 hour/week during pregnancy.

^bSequence 1 includes the Play, Still-Face 1 and Reunion 1 episodes; Sequence 2 includes the Still-Face 2 and Reunion 2 episodes.

Table S4. Results from sensitivity analyses examining change (β , 95% CI) in RSA_c at baseline (Play) and between sequential SFP-R episodes (Δ RSA_c) for a 1-unit increase in PM_{2.5} ($\mu\text{g}/\text{m}^3$)

	All (n=237)	Both SFP-R sequences completed (n=185)	Participants with a SF violation excluded (n=191)	Infants with a chronic medical condition excluded (n=223)	Infants not alert and rested at testing excluded (n=207)	Infants with postnatal exposure to ETS excluded (n=223)
Model 1: Play ^a	-3.53 (-6.96, 0.02)	-3.05 (-6.94, 0.98)	-3.80 (-7.49, 0.03)	-3.42 (-6.88, 0.16)	-3.79 (-7.49, 0.05)	-3.92 (-7.49, -0.20)
Model 2: Δ RSA _c Play to SF ₁	0.74 (-0.12, 1.60)	0.80 (-0.13, 1.74)	0.97 (0.06, 1.89)	0.68 (-0.19, 1.55)	0.82 (-0.12, 1.76)	0.89 (-0.01, 1.78)
Model 3: Δ RSA _c SF ₁ to R ₁	0.04 (-0.84, 0.92)	-0.25 (-1.19, 0.69)	-0.08 (-1.01, 0.84)	0.11 (-0.80, 1.01)	0.06 (-0.86, 0.98)	-0.27 (-1.11, 0.57)
Model 4: Δ RSA _c R ₁ to SF ₂	0.61 (-0.43, 1.66) ^b	0.61 (-0.43, 1.66) ^b	0.59 (-0.46, 1.64) ^c	0.60 (-0.48, 1.68) ^d	0.55 (-0.52, 1.62) ^e	0.89 (-0.15, 1.93) ^f
Model 5: Δ RSA _c SF ₂ to R ₂	-1.04 (-2.24, 0.15) ^b	-1.04 (-2.24, 0.15) ^b	-1.09 (-2.38, 0.20) ^c	-1.07 (-2.31, 0.16) ^d	-0.89 (-2.17, 0.38) ^e	-0.95 (-2.20, 0.30) ^f

Abbreviations: ETS: environmental tobacco smoke; PM_{2.5}: fine particulate matter; PRISM: PRogramming of Intergenerational Stress Mechanisms; RSA_c: respiratory sinus arrhythmia (in ms/ml corrected for T_{TOT}); SFP-R: Repeated Still-Face Paradigm; SF: Still-Face Adjusted for: infant activity, maternal race/ethnicity, maternal stress and material hardship

^aRSA_c at Play is natural-log transformed, therefore estimates are interpreted as the percent change in RSA_c for a 1-unit increase in PM_{2.5}

^bn=185, estimates for the second SFP-R sequence (R1-SF2-R2) are identical for the overall model and the model subset on infants completing both SFP-R sequences because the underlying subset of infants is the same.

^cn=135; ^dn=174; ^en=162; ^fn=177

Table S5. Results from sensitivity analyses examining change (β , 95% CI) in RSA_c at baseline (Play) and between sequential SFP-R episodes (Δ RSA_c) for a 1-unit increase in prenatal PM_{2.5} ($\mu\text{g}/\text{m}^3$) adjusting for postnatal PM_{2.5} exposure.

	Prenatal PM _{2.5} (n=237)	Prenatal PM _{2.5} subset on infants with postnatal PM _{2.5} measures (n=204)	Adjusted for chronic ^b postnatal PM _{2.5} (n=204)	Adjusted for short-term ^c postnatal PM _{2.5} (n=204)
Model 1: Play ^a	-3.53 (-6.96, 0.02)	-3.81 (-7.56, -0.05)	-3.93 (-8.35, 0.50)	-3.70 (-7.46, 0.00)
Model 2: Δ RSA _c Play to SF ₁	0.74 (-0.12, 1.60)	0.68 (-0.22, 1.58)	0.63 (-0.43, 1.70)	0.67 (-0.23, 1.57)
Model 3: Δ RSA _c SF ₁ to R ₁	0.04 (-0.84, 0.92)	0.05 (-0.87, 0.97)	0.41 (-0.67, 1.49)	0.10 (-0.82, 1.02)
Model 4: Δ RSA _c R ₁ to SF ₂	0.61 (-0.43, 1.66)	0.83 (-0.31, 1.97)	0.31 (-1.05, 1.67) ^d	0.76 (-0.39, 1.92) ^d
Model 5: Δ RSA _c SF ₂ to R ₂	-1.04 (-2.24, 0.15)	-1.40 (-2.60, -0.19)	-1.26 (-2.70, 0.18) ^d	-1.57 (-2.78, -0.35) ^d
Abbreviations: PM _{2.5} : fine particulate matter; RSA _c : respiratory sinus arrhythmia (in ms/ml corrected for T _{TOT}); SFP-R: Repeated Still-Face Paradigm All models adjusted for: infant activity, maternal race/ethnicity, maternal stress and material hardship ^a RSA _c at Play is natural-log transformed, therefore estimates are interpreted as the percent change in RSA _c for a 1-unit increase in PM _{2.5} ^b Average modeled residential PM _{2.5} exposure between birth and date of SFP-R testing at approximately 6 months of age ^c Modeled residential PM _{2.5} exposure on the day of SFP-R testing ^d n=158				

Table S6. Adjusted change in RSA_c between sequential episodes per a 1-unit increase in PM_{2.5} ($\mu\text{g}/\text{m}^3$) in models adjusting for laboratory conditions on the day of SFP-R testing

	Play ^a	Δ RSA _c Play to SF ₁	Δ RSA _c SF ₁ to R ₁	Δ RSA _c R ₁ to SF ₂	Δ RSA _c SF ₂ to R ₂
Main model (n=237)	-3.53 (-6.96, 0.02)	0.74 (-0.12, 1.60)	0.04 (-0.84, 0.92)	0.61 (-0.43, 1.66) ^b	-1.04 (-2.24, 0.15) ^b
Main model (n=142)	-3.86 (-8.41, 0.92)	0.87 (-0.31, 2.06)	-0.04 (-1.23, 1.15)	0.53 (-0.91, 1.97) ^c	-1.44 (-2.95, 0.06) ^c
+ lab temperature (n=142)	-3.81 (-8.45, 1.07)	0.89 (-0.32, 2.09)	-0.05 (-1.26, 1.16)	0.52 (-0.95, 1.98) ^c	-1.48 (-3.01, 0.04) ^c
+ lab humidity (n=142)	-3.85 (-8.42, 0.94)	0.83 (-0.35, 2.02)	-0.07 (-1.27, 1.12)	0.58 (-0.86, 2.01) ^c	-1.41 (-2.91, 0.10) ^c
+ lab barometric pressure (n=142)	-3.87 (-8.42, 0.90)	0.88 (-0.30, 2.06)	-0.05 (-1.23, 1.13)	0.53 (-0.91, 1.97) ^c	-1.45 (-2.95, 0.06) ^c
+ travel time to lab (n=142)	-3.86 (-8.41, 0.92)	0.87 (-0.31, 2.06)	-0.04 (-1.22, 1.14)	0.54 (-0.89, 1.96) ^c	-1.44 (-2.95, 0.06) ^c
Abbreviations: PM _{2.5} : fine particulate matter; RSA _c : respiratory sinus arrhythmia (in ms/ml corrected for T _{TOT}); SFP-R: Repeated Still-Face Paradigm All models adjusted for: infant activity, maternal race/ethnicity, maternal stress and material hardship ^a RSA _c at Play is natural-log transformed, therefore estimates are interpreted as the percent change in RSA _c for a 1-unit increase in PM _{2.5} ^b N=185; ^c N=111					

Table S7. Sensitivity analyses examining change in RSA _c at baseline and between SFP-R episodes for a 1-unit increase in PM _{2.5} in models adjusting for maternal LSC-R scores treated as a dichotomous versus continuous variable (n=237)		
	Adjust for LSC-R as a continuous variable	Adjust for LSC-R as a dichotomous variable ^a
Model 1: Play	-3.53 (-6.96, 0.02)	-3.66 (-7.08, -0.12)
Model 2: Δ RSA _c Play to SF ₁	0.74 (-0.12, 1.60)	0.76 (-0.09, 1.62)
Model 3: Δ RSA _c SF ₁ to R ₁	0.04 (-0.84, 0.92)	0.11 (-0.77, 0.98)
Model 4: Δ RSA _c R ₁ to SF ₂	0.61 (-0.43, 1.66) ^c	0.56 (-0.48, 1.59) ^c
Model 5: Δ RSA _c SF ₂ to R ₂	-1.04 (-2.24, 0.15) ^c	-1.16 (-2.35, 0.04) ^c
Abbreviations: LSC-R: Life Stressor Checklist – Revised; PM_{2.5}: fine particulate matter; RSA_c: respiratory sinus arrhythmia (in ms/ml corrected for T_{TOT}); SFP-R: Repeated Still-Face Paradigm All models adjusted for: infant activity, maternal race/ethnicity, maternal stress (LSC-R) and material hardship ^aStratified at the 4th quartile (>11 versus ≤ 11) ^bRSA_c at Play is natural-log transformed, therefore estimates are interpreted as the percent change in RSA_c for a 1-unit increase in PM_{2.5} ^cN=185		

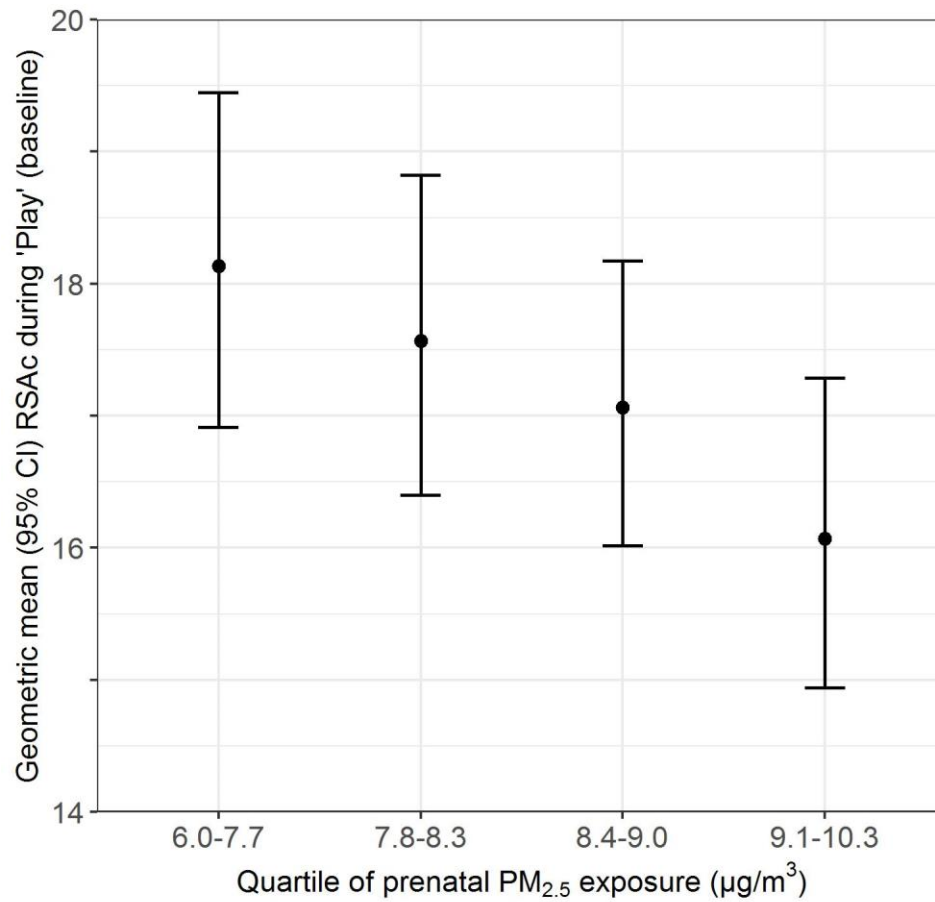


Figure S1. Geometric mean RSA_c at the baseline (Play) SFP-R episode by quartile of prenatal PM_{2.5} exposure among 237 infants enrolled in the Boston-based PRISM cohort.

Abbreviations: PM_{2.5}: fine particulate matter; PRISM: PRogramming of Intergenerational Stress Mechanisms; R: Reunion; RSA_c: respiratory sinus arrhythmia in units ms/ml corrected for total respiratory cycle time (T_{TOT}); SF: Still-Face; SFP-R: Repeated Still-Face Paradigm.

Values are least square geometric means estimated using linear regression adjusting for infant activity during the Play episode, race/ethnicity, maternal stress, and material hardship.

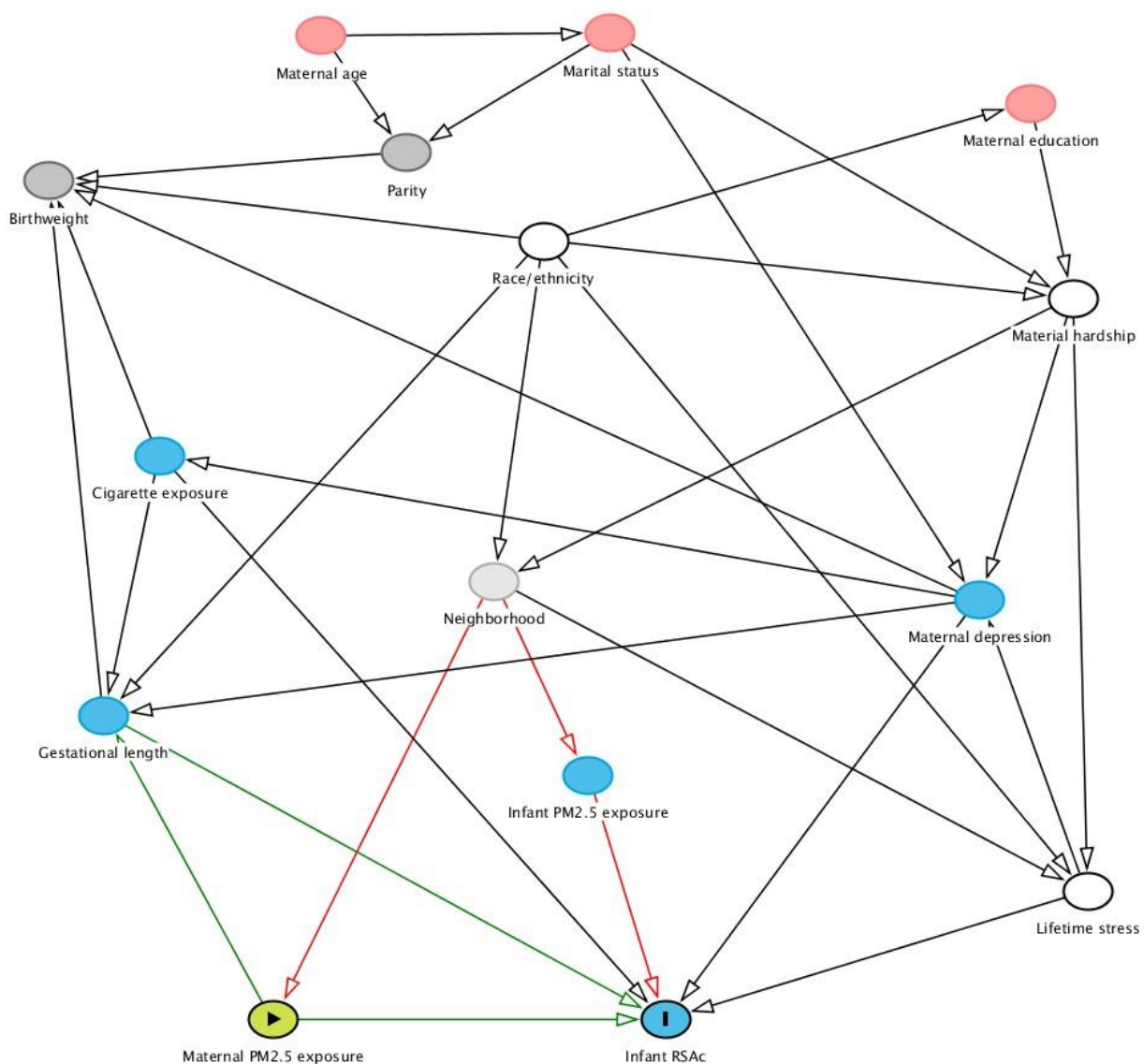


Figure S2. Directed Acyclic Graph illustrating implied conditional dependencies between study variables.

Abbreviations: PM_{2.5}: fine particulate matter, RSA_c: respiratory sinus arrhythmia

Green lines represent unbiased causal paths. Red lines represent potentially biased paths. The green circle with the triangle represents the exposure of interest (maternal PM_{2.5}). The blue circle with a vertical line represents the outcome of interest (RSA_c). White circles represent adjusted variables in the minimally sufficient adjustment set (maternal stress, race/ethnicity, material hardship). Sensitivity analyses investigate models additionally adjusting for infant PM_{2.5} exposure (Table S5). Diagram generated using DAGitty v2.3 available at: <http://www.dagitty.net/dags.html#>

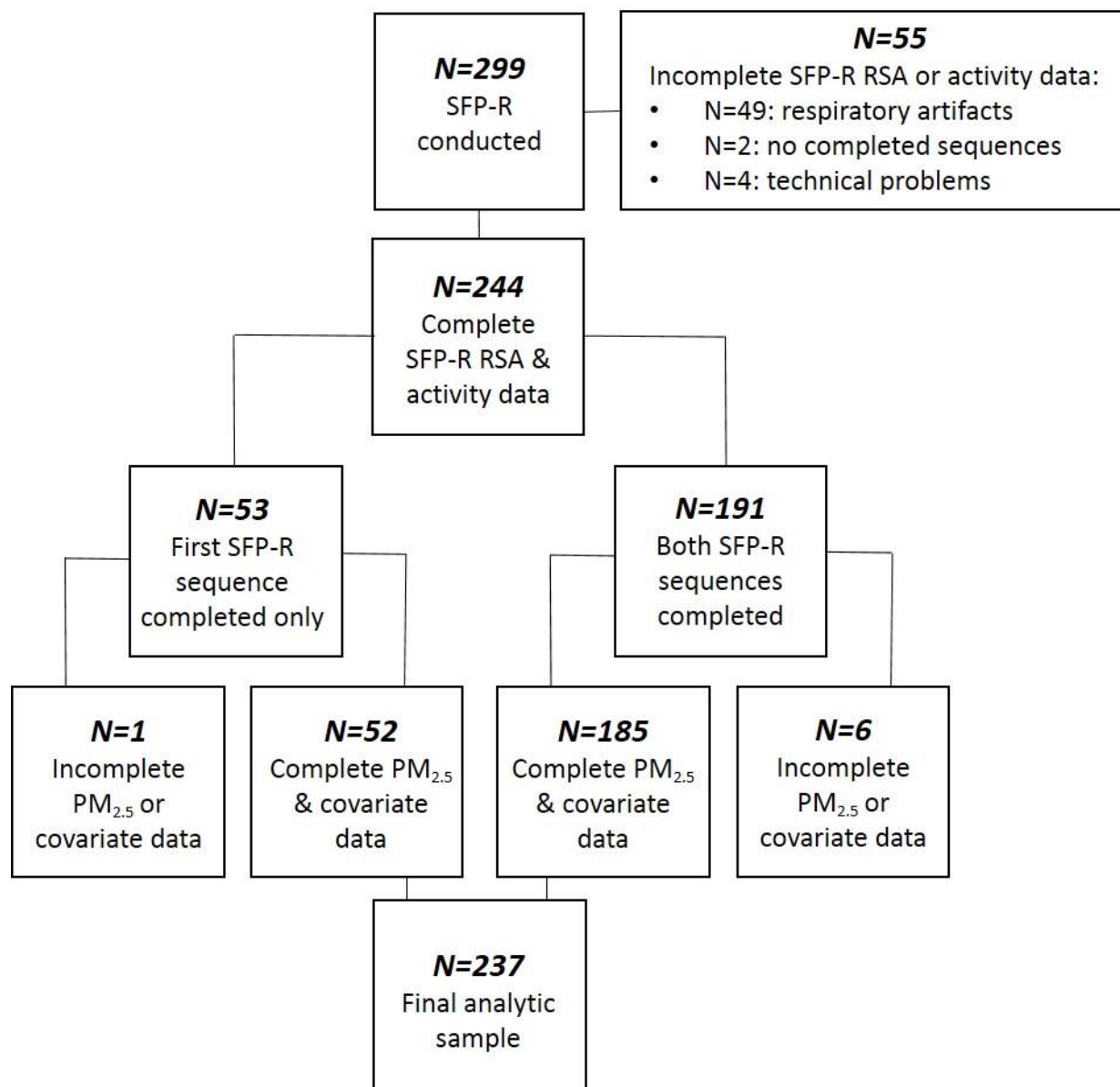


Figure S3. Schematic outlining selection of the analytic sample from the underlying PRISM cohort